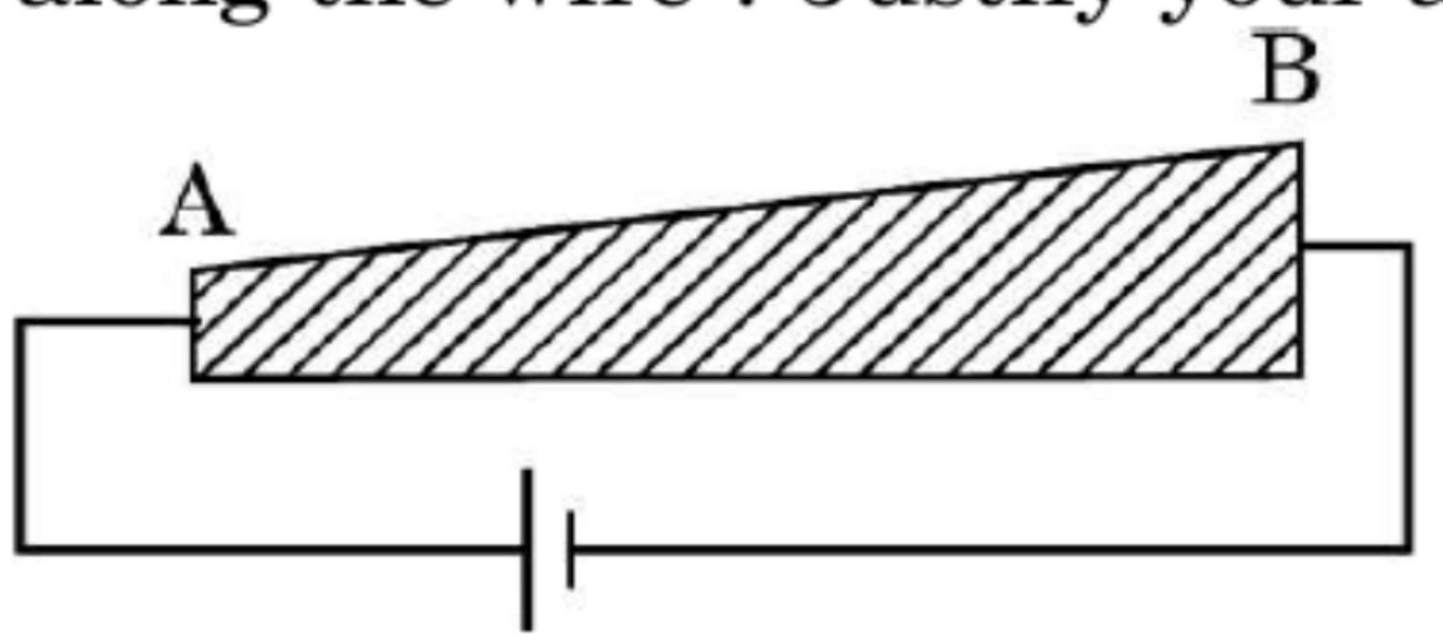
CBSE 2023

2. A current flows through a series combination of two copper wires of equal length but their radii are in the ratio of 1 : 2. The ratio of drift velocities of free electrons in the wires will be :

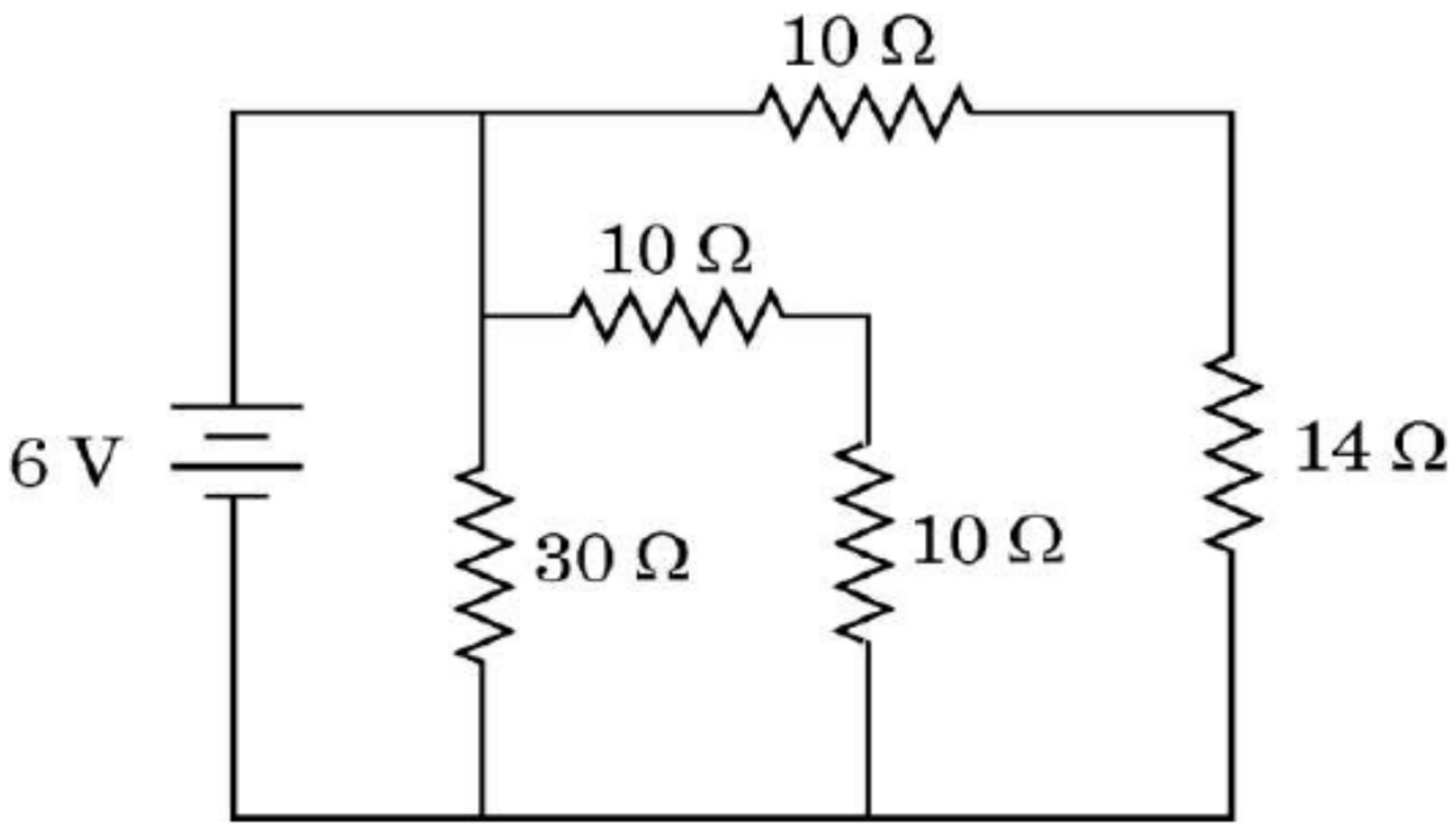
(a) 8 (b) 4 (c) 2 (d) 1

32. (a) (i) Define mobility of electrons. Give its SI units.

(ii) A steady current flows through a wire AB, as shown in the figure. What happens to the electric field and the drift velocity along the wire ? Justify your answer.



(iii) Consider the circuit shown in the figure. Find the effective resistance of the circuit and the current drawn from the battery. *5*



8. The SI unit of mobility of charge carriers is :

(a) $\Omega\ s^{-1}$

(b) $m^2\ V^{-1}\ s^{-1}$

(c) $m\ s^{-1}\ V^{-1}$

(d) $\Omega\ m$

3. The current density due to drift of electrons in a conductor is given by (symbols have their usual meanings)

(a) $n\,e\,A\,v_d$

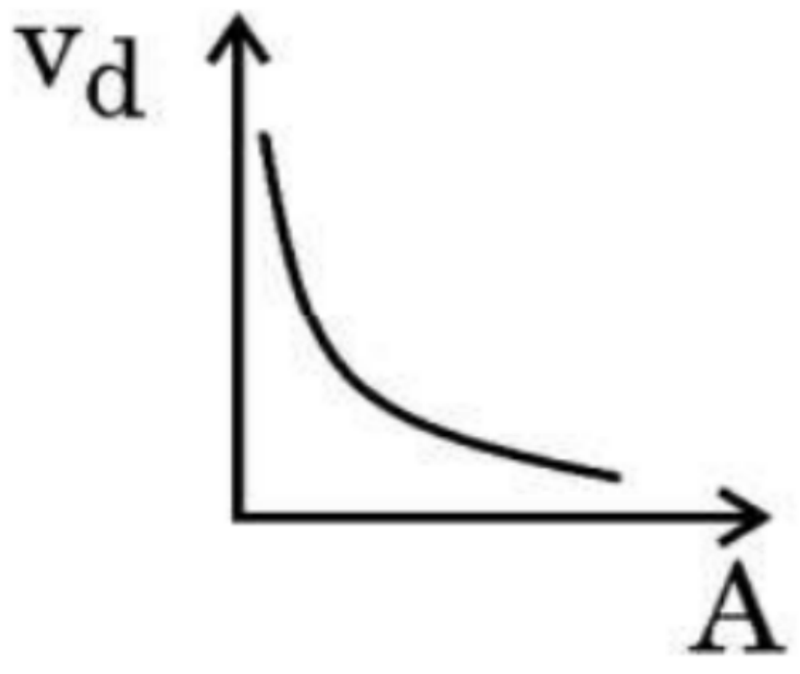
(b) $\dfrac{n\,A\,v_d}{e}$

(c) $\dfrac{n\,v_d}{e\,A}$

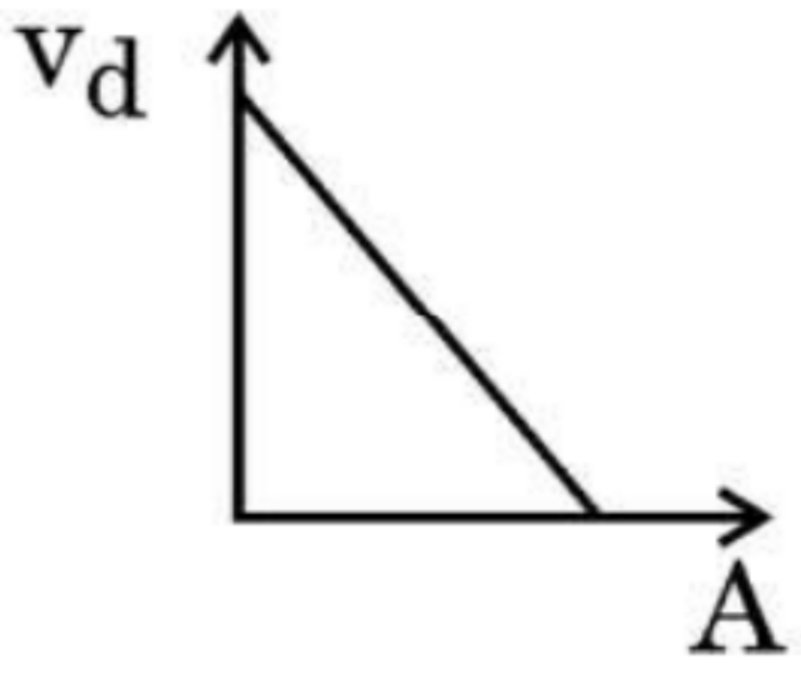
(d) $n\,e\,v_d$

26. A potential difference V is applied across a conductor of length l and uniform cross-section area A. How will the (i) electric field E, (ii) drift velocity v_d, and (iii) current density j be affected when (a) V is doubled and (b) l is halved (keeping other factors constant) ? *3*

2. A steady current flows through a metallic wire whose area of cross-section (A) increases continuously from one end of the wire to the other. The magnitude of drift velocity (v_d) of the free electrons as a function of 'A' can be shown by :



(a)



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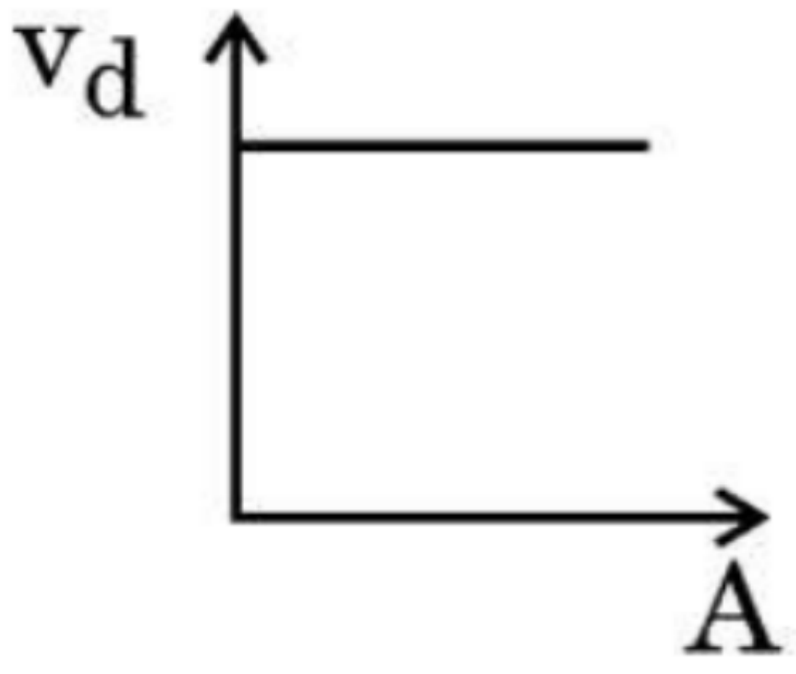
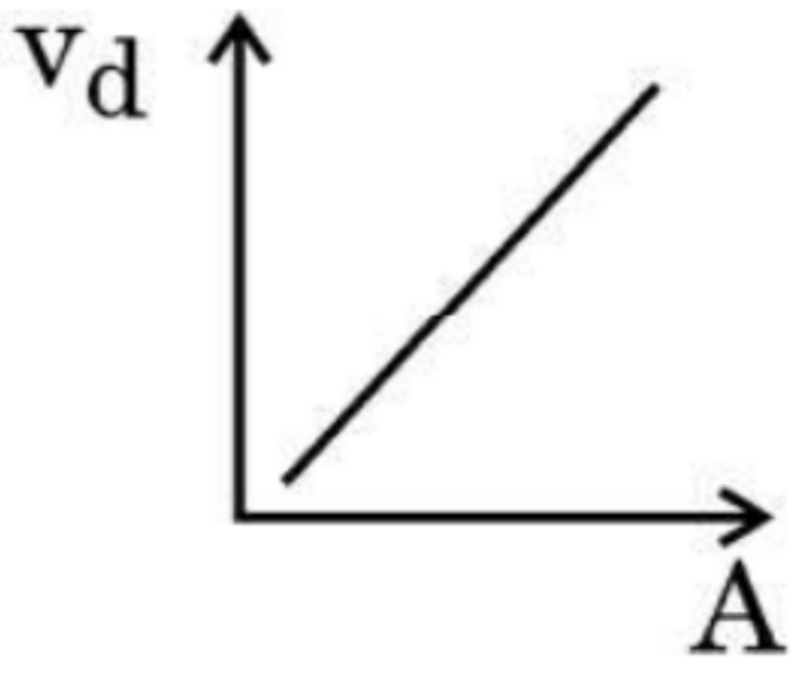


(c)



(d)

11. The current in a device varies with time t as $I = 6\,t$, where I is in mA and t is in s. The amount of charge that passes through the device during $t = 0s$ to $t = 3s$ is **1**

(a) 10 mC (b) 18 mC

(c) 27 mC (d) 54 mC

29. A potential difference V is applied across a conductor of length l and cross-sectional area A. Briefly explain how the current density j in the conductor will be affected if

(a) the potential difference V is doubled,

(b) the conductor were gradually stretched to reduce its cross-sectional area to $\frac{A}{2}$ and then the same potential difference V is applied across it. **3**

7. A steady current of 8 mA flows through a wire. The number of electrons passing through a cross-section of the wire in 10 s is **1**

(a) 4.0×10^{16} (b) 5.0×10^{17}

(c) 1.6×10^{16} (d) 1.0×10^{17}

27. Define current density and relaxation time. Derive an expression for resistivity of a conductor in terms of number density of charge carriers in the conductor and relaxation time. **3**

24. Two conductors, made of the same material have equal lengths but different cross-sectional areas A_1 and A_2 ($A_1 > A_2$). They are connected in parallel across a cell. Show that the drift velocities of electrons in two conductors are equal. **2**

33. (a) (i) Explain how free electrons in a metal at constant temperature attain an average velocity under the action of an electric field. Hence obtain an expression for it. **3**

(ii) Consider two conducting wires A and B of the same diameter but made of different materials joined in series across a battery. The number density of electrons in A is 1.5 times that in B. Find the ratio of drift velocity of electrons in wire A to that in wire B. **2**

24. A potential difference (V) is applied across a conductor of length ‘L’ and cross-sectional area ‘A’.

How will the drift velocity of electrons and the current density be affected if another identical conductor of the same material were connected in series with the first conductor ? Justify your answers. **2**

24. The potential difference applied across a given conductor is doubled. How will this affect (i) the mobility of electrons and (ii) the current density in the conductor ? Justify your answers. **2**

26. A cylindrical conductor of resistance R is connected to a battery. The drift velocity of electrons in this conductor is v_d. The conductor is disconnected from the battery and gradually stretched so that its length increases by 25%. It is again connected to the same battery. Find the new value of : *3*

(a) the drift speed of electrons, and

(b) the resistance of the conductor.

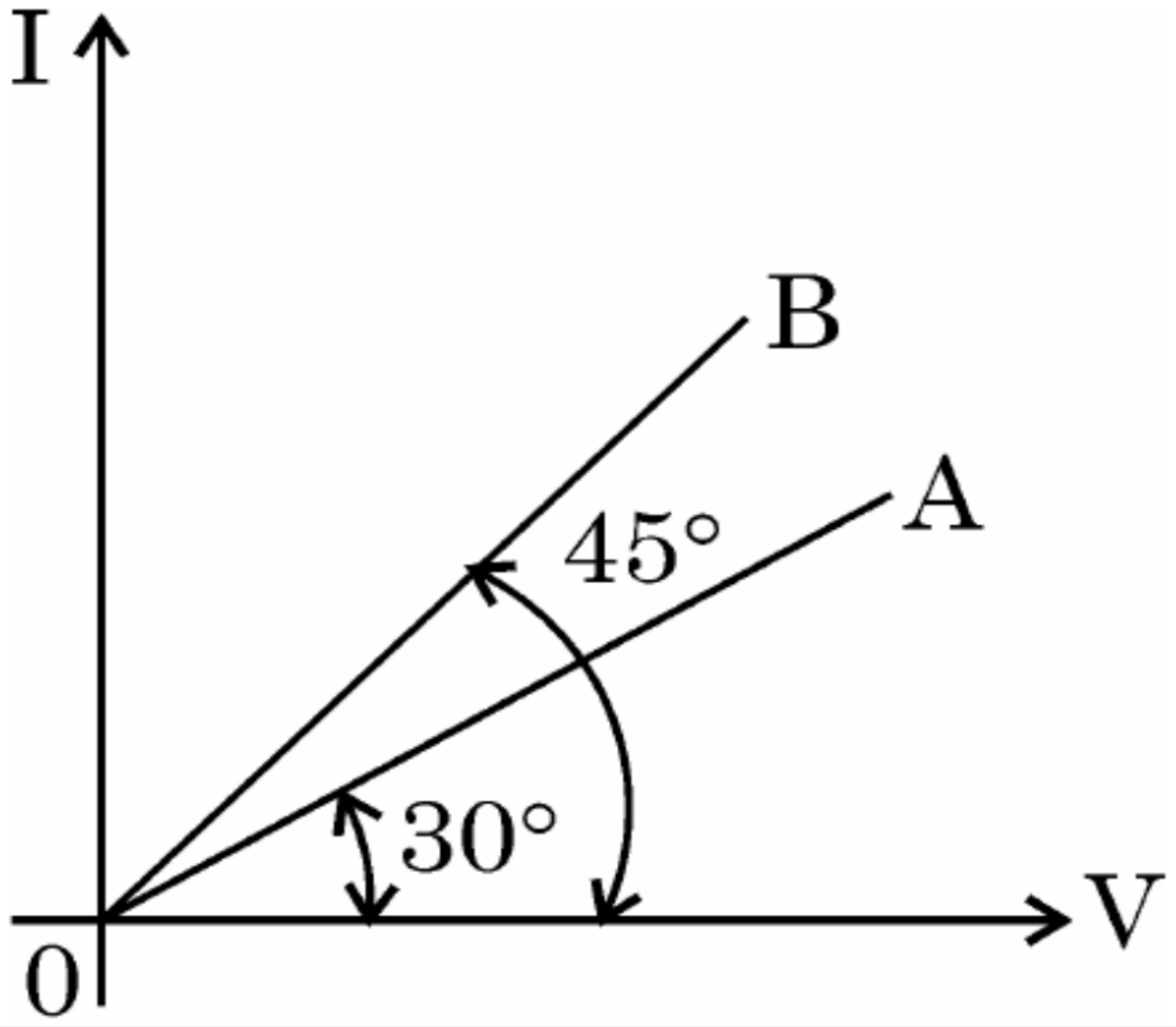
28. A potential difference of 1.0 V is applied across a conductor of length 5.0 m and area of cross-section 1.0 mm^2. When current of 4.25 A is passed through the conductor, calculate

(i) the drift speed and (ii) relaxation time, of electrons. (Given number density of electrons in the conductor, $n = 8.5 \times 10^{28}\ m^{-3}$).

20. Two wires A and B of different metals have their lengths in ratio 1 : 2 and their radii in ratio 2 : 1 respectively. I-V graphs for them is shown in the figure. Find the ratio of their **2**

(i) Resistances (R_A/R_B) and

(ii) Resistivities (σ_A/σ_B)



17. (a) What is meant by 'relaxation time' of free electrons in a conductor ?

Show that the resistance of a conductor can be expressed by $R = \dfrac{m\,l}{ne^2\tau\,A}$, where symbols have their usual meanings. *2*

OR

(b) Draw the circuit diagram of a Wheatstone bridge. Obtain the condition when no current flows through the galvanometer in it. *2*

22. A current of 1·6 A flows through a wire when a potential difference of 1·0 V is applied across it. The length and cross-sectional area of the wire are 1·0 m and $1{\cdot}0 \times 10^{-7}\ m^2$ respectively. Calculate : *3*

(a) Electric field across the wire

(b) Current density

(c) Average relaxation time (τ)

(Number density of free electrons in the wire is $9{\cdot}0 \times 10^{28}\ m^{-3}$)

2. Electrons drift with speed v_d in a conductor with potential difference V across its ends. If V is reduced to $\left(\frac{V}{2}\right)$, their drift speed will become :

(A) $\frac{v_d}{2}$

(B) v_d

(C) $2\ v_d$

(D) $4\ v_d$

17. An electric field E is maintained in a wire of length 'l' and area of cross-section 'a'. Derive the relation between the current density 'σ' in the wire and the electric field E. *2*

17. (a) "The electron drift speed is only a few mm/s for currents in the range of a few amperes for a given conductor." How then is current established almost the instant a circuit is closed ? Explain.

(b) 'V = IR is a statement of Ohm's Law' is not true. Explain. *2*

17. Define resistivity of a conductor. How does the resistivity of a conductor depend upon the following : *2*

(a) Number density of free electrons in the conductor (n)

(b) Their relaxation time (τ)

3. In a uniform straight wire, conduction electrons move along + x direction. Let $\vec{E}$ and $\vec{j}$ be the electric field and current density in the wire, respectively. Then :

(A) $\vec{E}$ and $\vec{j}$ both are along + x direction.

(B) $\vec{E}$ and $\vec{j}$ both are along – x direction.

(C) $\vec{E}$ is along + x direction, but $\vec{j}$ is along – x direction.

(D) $\vec{E}$ is along – x direction, but $\vec{j}$ is along + x direction.

17. A uniform wire of length L and area of cross-section A has resistance R. The wire is uniformly stretched so that its length increases by 25%. Calculate the percentage increase in the resistance of the wire. *2*

23. (a) Define current density. Is it a scalar or a vector ? An electric field $\vec{E}$ is maintained in a metallic conductor. If n be the number of electrons (mass m, charge $-e$) per unit volume in the conductor and τ its relaxation time, show that the current density $\vec{j} = \alpha \vec{E}$, where $\alpha = \left(\dfrac{ne^2}{m}\right)\tau$. *3*

OR

(b) What is a Wheatstone bridge ? Obtain the necessary conditions under which the Wheatstone bridge is balanced. *3*

22. (a) Define the terms – 'temperature coefficient of resistance' and 'electrical conductivity' of a conductor. Why are constantan and manganin used for making standard resistances ? Explain. *3*

OR

(b) Define 'drift velocity'. Show that resistivity of the material of a conductor is inversely proportional to the relaxation time for the free electrons in the conductor. *3*

15. *Assertion (A) :* The current density ($\vec{J}$) at a point in a conducting wire is in the direction of electric field ($\vec{E}$) at that point.

Reason (R) : A conducting wire obeys Ohm's law.

28. (a) The radius of a conducting wire AB uniformly decreases from its one end A to another end B. It is connected across a battery. How will (i) electric field, (ii) current density, and (iii) mobility of electrons change from end A to end B ? Justify your answer in each case. *3*

17. Two conducting wires of same material have radii r and $\frac{r}{2}$. The current flowing through them is I and 2I, respectively. Find the ratio of drift velocity of free electrons in the wires. *2*

29. In a metallic conductor, an electron, moving due to thermal motion, suffers collisions with the heavy fixed ions but after collision, it will emerge out with the same speed but in random directions. If we consider all the electrons, their average velocity will be zero. When an electric field is applied, electrons move with an average velocity, known as drift velocity (v_d). The average time between successive collisions is known as relaxation time (τ). The magnitude of drift velocity per unit electric field is called mobility (μ).

An expression for current through the conductor can be obtained in terms of drift velocity, number of electrons per unit volume (n), electronic charge (– e), and the cross-sectional area (A) of the conductor. This expression leads to an expression between current density ($\hat{j}$) and the electric field ($\vec{E}$). Hence, an expression for resistivity (ρ) of a metal is obtained. This expression helps us to understand increase in resistivity of a metal with increase in its temperature, in terms of change in the relaxation time (τ) and change in the number density of electrons (n).

(i) (a) Consider the contribution of the following two factors I and II in resistivity of a metal :

I. Relaxation time of electrons

II. Number of electrons per unit volume

The resistivity of a metal increases with increase in its temperature because :

(A) I decreases and II increases.

(B) I increases and II is almost constant.

(C) Both I and II increase.

(D) I decreases and II is almost constant.

OR

(b) A steady current flows in a copper wire of non-uniform cross-section. Consider the following three physical quantities :

I. Electric field

II. Current density

III. Drift speed

Then at the different points along the wire : *1*

(A) II and III change, but I is constant.

(B) I and II change, but III is constant.

(C) I and III change, but II is constant.

(D) All I, II and III change.

(iii) A wire of length 0·5 m and cross-sectional area $1\cdot0 \times 10^{-7}$ m^2 is connected to a battery of 2 V that maintains a current of 1·5 A in it. The conductivity of the material of the wire (in Ω^{-1} m^{-1}) is : *1*

(A) $2\cdot5 \times 10^4$

(B) $3\cdot0 \times 10^5$

(C) $3\cdot75 \times 10^6$

(D) $5\cdot0 \times 10^7$

(iv) Consider two cylindrical conductors A and B, made of the same metal connected in series to a battery. The length and the radius of B are twice that of A. If μ_A and μ_B are the mobility of electrons in A and B respectively, then $\frac{\mu_A}{\mu_B}$ is : *1*

(A) $\frac{1}{2}$

(B) $\frac{1}{4}$

(C) 2

(D) 1

17. Two wires made of the same material have the same length (l) but different cross-sectional areas A_1 and A_2. They are connected together with a cell of voltage V. Find the ratio of the drift velocities of free electrons in the two wires when they are joined in (i) series, and (ii) parallel. *2*

18. (a) What is the difference between 'velocity' and 'drift velocity' of electrons in a current-carrying conductor.

(b) A copper wire of uniform cross-sectional area carries a current of 3·4 A. The drift velocity of conduction electrons is 0·2 mm/s. If the number density of electrons in copper is $8{\cdot}5 \times 10^{28}$ m^{-3}, find the area of cross-section of the wire. *2*

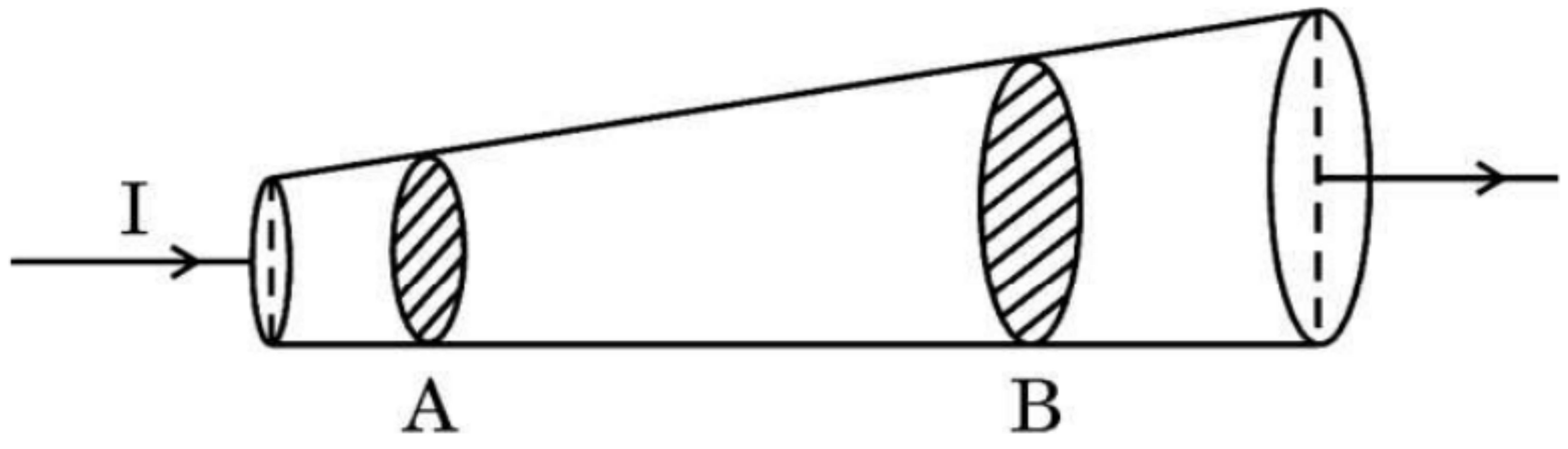
22. (a) Define the term 'drift velocity' of conduction electrons in a conductor.

(b) A conductor of length l and area of cross-section A is connected across an ideal battery of emf V. Derive the formula for the current density in terms of relaxation time τ. *3*

17. A 20.0 cm long wire is connected across an ideal battery of 4 V. The drift velocity of conduction electrons in the wire is 0.8 mm/s. Calculate the relaxation time of the electrons. **2**

(b) Define the term current density. Write its SI unit. Derive the equivalent form of Ohm's law $\vec{j} = \sigma\vec{E}$. **3**

(b) (i) Current I (= 1 A) is passing through a copper rod ($n = 8 \cdot 5 \times 10^{28}\ m^{-3}$) of varying cross-sections as shown in the figure. The areas of cross-section at points A and B along its length are $1 \cdot 0 \times 10^{-7}\ m^2$ and $2 \cdot 0 \times 10^{-7}\ m^2$ respectively. Calculate :



(I) the ratio of electric fields at points A and B.

(II) the drift velocity of free electrons at point B.

18. A wire of resistance X ohm is gradually stretched till its length becomes twice its original length. If its new resistance becomes 40 Ω, find the value of X. *2*

2. Three wires A, B and C of the same material have lengths and area of cross-sections as $(2l, \frac{A}{2})$, (l, A) and $(\frac{l}{2}, 2A)$, respectively. If the resistances of these wires are R_A, R_B and R_C respectively, then :

(A) $R_A > R_B > R_C$

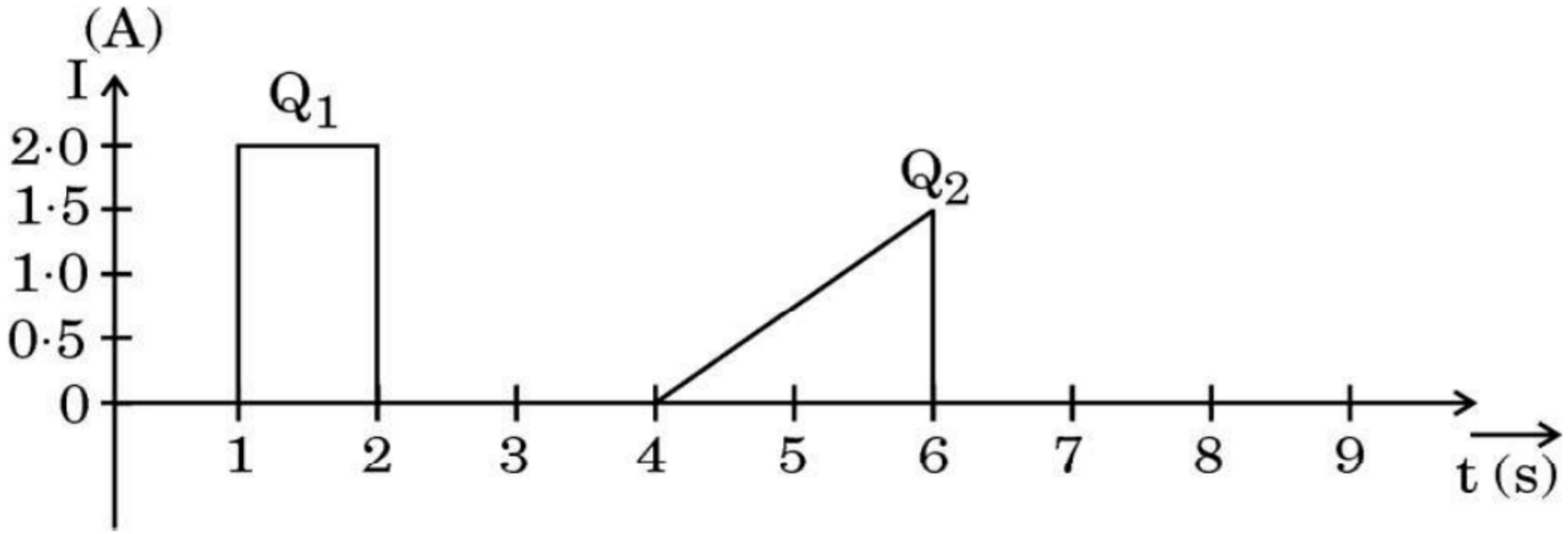
(B) $R_B > R_C > R_A$

(C) $R_C > R_A > R_B$

(D) $R_A > R_C > R_B$

27. (a) (i) Derive an expression for the resistivity of a conductor in terms of number density of free electrons and relaxation time.

(ii) The figure shows the plot of current through a cross-section of wire over two different time intervals. Compare the charges (Q_1 and Q_2) that pass through the cross-section during these time intervals. *3*



2. When the switch of the circuit is turned on, the filament of the bulb glows instantaneously because :

(A) the electrons coming from the power source move fast through the initially empty filament.

(B) the filament may be old having low resistance.

(C) electric field is established instantaneously across the filament which pushes the electrons.

(D) free electrons in the filament travel with the speed of light.

2. A current flows through a cylindrical conductor of radius R. The current density at a point in the conductor is $j = \alpha r$ (along its axis), here α is a constant and r is distance from the axis of the conductor. The current flowing through the portion of the conductor from $r = 0$ to $r = \frac{R}{2}$ is proportional to :

(A) R

(B) R^2

(C) R^3

(D) R^4

22. (a) Define Electrical conductivity. Obtain the expression of electrical conductivity of a conductor in terms of number density and relaxation time of free electrons. **3**

(b) Explain qualitative change in resistivity of a conductor with temperature using expression obtained in (a).

2. Two conductors A and B of the same material have their lengths in the ratio 1:2 and radii in the ratio 2:3. If they are connected in parallel across a battery, the ratio $\frac{v_A}{v_B}$ of the drift velocities of electrons in them will be - **1**

(A) 2 (B) $\frac{1}{2}$

(C) $\frac{3}{2}$ (D) $\frac{8}{9}$

(b) (i) A conductor of length l is connected across an ideal cell of emf E. Keeping the cell connected, the length of the conductor is increased to $2l$ by gradually stretching it. If R and R' are initial and final values of resistance and ν_d and $\nu_d{}'$ are initial and final values of drift velocity, find the relation between (i) R' and R and (ii) $\nu_d{}'$ and ν_d.

(ii) When electrons drift in a conductor from lower to higher potential, does it mean that all the 'free electrons' of the conductor are moving in the same direction ?

2. The resistance of a wire of length L and radius r is R. Which one of the following would provide a wire of the same material of resistance $\frac{R}{2}$? **1**

(A) Using a wire of same radius and twice the length
(B) Using a wire of same radius and half length
(C) Using a wire of same length and twice the radius
(D) Using a wire of same length and half the radius

17. Show that $\vec{E} = \rho \vec{j}$ leads to Ohm's law. Write a condition in which the Ohm's law is not valid for a material. **2**